IS1212 : Probability and Statistics

K S Saubhagya
Lecturer (Probationary)
Department of Statistics, Faculty of Science, University of Colombo
saubhagya@stat.cmb.ac.lk

The Normal Distribution

The normal distribution is the most important continuous distribution in statistics. Many measured quantities in the natural sciences follow a normal distribution. Eg. heights, masses, ages, examination results.

p.d.f of Normal Variable

A continuous random variable X having p.d.f

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

is said to have a normal distribution with mean μ and variance σ^2.

- μ and σ^2 are the parameters of the distribution.
- The distribution is bell shaped and symmetrical about $x = \mu$.

Expectation and Variance

If $X \sim N(\mu, \sigma^2)$ then $E(X) = \mu$ and $Var(X) = \sigma^2$.

Approximately 95% of the distribution lies within ±2 standard deviations of the mean .

Approximately 99.8% of the distribution lies within ±3 standard deviations of the mean .

The Standard Normal Distribution

Consider the random variable Z where $Z = \frac{X - \mu}{\sigma}$

Now if the random variable X has a normal distribution with mean μ and variance σ^2, then the random variable Z has a standard normal distribution with mean 0 and variance 1.

The p.d.f of Z, Ø(X)

The p.d.f. of the standard normal variable X is denoted by Ø(X) where

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}.$$

The cumulative distribution function for X

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-t^2/2}\, dt$$

To obtain this value easily, we refer to standard Normal Tables.

Exercise 1

If $Z \sim N(0,1)$, find from tables

(a) $P(Z < 1.377)$

(b) $P(Z > -1.377)$

(c) $P(Z > 1.377)$

(d) $P(Z < -1.377)$

Exercise 2

If $Z \sim N(0,1)$, find from tables

(a) $P(0.345 < Z < 1.751)$

(b) $P(-1.4 < Z < -0.6)$

Exercise 3

The r.v $X \sim N(300,25)$. Find

(a) $P(X>305)$

(b) $P(X<291)$

Exercise 3 - Solution

The r.v $X \sim N(300,25)$.

(a) $P\left(\frac{X-\mu}{\sigma} > \frac{305-300}{5}\right)$

$P(Z>1) = 1-P(Z<1) = 0.1587$

(b) $P(X<291) = P\left(Z<\frac{291-300}{5}\right) = P(Z< -9/5)$ (get the value from the table)

Exercise 4

The time taken by a milkman to deliver milk to the High street is normally distributed with mean 12 minutes and standard deviation 2 minutes. He delivers milk every day. Estimate the number of days during the year when he takes

(a) longer than 17 minutes

(b) less than 10 minutes

(c) between 9 and 13 minutes.

Solution 4

Let X be the r.v " the time taken to deliver milk to the high street"

$X \sim N(12, 4)$

(a) $P(X > 17) = P\left(Z > \frac{17-12}{2}\right)$

$= P(Z > 2.5)$

$= 1 - 0.99379$

$= 0.00621$

Number of days he takes longer than 17 minutes = 365 * 0.00621

=2.27 =2 days

(b) $P(X < 10) = P\left(Z < \frac{10-12}{2}\right)$

$= P(Z < -1)$

$= 1 - 0.8413$

$= 0.1587$

Number of days he takes les than 10 minutes = 365 * 0.1587

=57.9 =58 days

(c) $P(9 < X < 13) = P\left(\frac{9-12}{2} < Z < \frac{13-12}{2}\right)$

$= P(-1.5 < Z < 0.5)$

$=0.6247$

Number of days he takes between 9 and 13 minutes = 365 * 0.6247

=228 days

Exercise 5 & 6

- The lengths of certain items follow a normal distribution with mean μ cm and standard deviation 6cm. It is known that 4.78% of the items have a length greater than 82cm. Find the value of the mean μ.

- If $X \sim N(100,36)$ and $P(X>a)= 0.1093$, find the value of a.

Solution 5

Length $X \sim N(\mu, 36)$

$$P(X > 82) = 0.0478$$

$$P\left(Z > \frac{82 - \mu}{6}\right) = 0.0478$$

$$P\left(Z < \frac{82 - \mu}{6}\right) = 1 - 0.0478 = 0.9522$$

$$\frac{82 - \mu}{6} = 1.667 \longrightarrow \mu = 72$$

Solution

(6) $P(X>a) = 0.1093$

$$P\left(Z>\frac{a-100}{6}\right) = 0.1093$$

$$P\left(Z<\frac{a-100}{6}\right) = 1- 0.1093 = 0.8907$$

Therefore $\frac{(a - 100)}{6} = 1.23$

$a = 107.38$

Sum and difference of two independent normal variables

If X and Y are two independent normal variables such that $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$

then $X+Y \sim N(\mu_1+\mu_2, \sigma_1^2 + \sigma_2^2)$

and $X-Y \sim N(\mu_1-\mu_2, \sigma_1^2 + \sigma_2^2)$

Example

If $X \sim N(60,16)$ and $Y \sim N(70,9)$, find

(a) $P(X+Y<140)$

(b) $P(120<X+Y<135)$

(c) $P(Y-X>7)$

Solution

If $X \sim N(60,16)$ and $Y \sim N(70,9)$, find

(a) $P(X+Y<140)$

$(X+Y) \sim N(60+70,16+9)$

i.e $R=X+Y \sim N(130,25)$

We require $P(R<140)=P(Z<2)$

$= 0.9772$

Solution

(b) $(X+Y) \sim N(60+70, 16+9)$

i.e $R = X+Y \sim N(130, 25)$

We require $P(120 < R < 135) = P(-2 < Z < 1)$

$= 0.8185$

Solution

(c) $(Y-X) \sim N(70-60, 9+16)$

i.e $T = Y-X \sim N(10, 25)$

We require $P(Y-X>7) = P(T>7)$

$= P(Z>-0.6)$

$= 0.7257$

Multiples of Normal Variables

If X is a normal variable such that $X \sim N(\mu, \sigma^2)$ then

$$aX \sim N(a\mu, a^2\sigma^2)$$

Example: If $X \sim N(50,25)$, find $P(3X>160)$

Solution

If $X \sim N(50,25)$, find $P(3X>160)$

$3X \sim N(150,225)$

We require $P(3X>160) = P(Z>0.667)$

$= 0.2523$

The Distribution of the Sample Mean

If X1, X2,.....Xn is a random sample of size n taken from a normal distribution with mean μ and variance σ^2 such that $X \sim N(\mu, \sigma^2)$, then the distribution of $\overline{X}$ is also normal with mean μ and variance σ^2/n.

Example

A random sample of size 15 is taken from a normal distribution with mean 60 and standard deviation 4. Find the probability that the mean of the sample is less than 58.

Solution

A random sample of size 15 is taken from a normal distribution with mean 60 and standard deviation 4. Find the probability that the mean of the sample is less than 58.

X~N(60,16)

Sample mean of size 15, $\overline{X}$ ~N(60, 16/15)

We require P($\overline{X}$ < 58) = P(Z<-1.936)= 0.0264

The Central Limit Theorem

The Distribution of the sample mean from any population

If X1, X2,.....Xn is a random sample of size n taken from **any** distribution with mean μ and variance σ^2 , **for large n**, then the distribution of sample mean is approximately normal with mean μ and variance σ^2/n.

Note: the approximation gets better as n gets larger

Example

If a random sample of size 30 is taken from the Binomial(9,0.5) distribution, find the probability that the sample mean exceeds 5.

Solution

If a random sample of size 30 is taken from the Binomial(9,0.5) distribution, find the probability that the sample mean exceeds 5.

X~Bin(9,0.5) np=4.5 npq=2.25

By CLT $\overline{X}$ ~N(4.5, 2.25/30)

P($\overline{X}$ >5) =P(Z>1.826) = 0.0340

Any questions?